

AVCulpability_Behavioral_Simulation (#260644)

Author(s)

This pre-registration is currently anonymous to enable blind peer-review.
It has 2 authors.

Pre-registered on:

2025/11/26 07:26 (PT)

1) Have any data been collected for this study already?

No, no data have been collected for this study yet.

2) What's the main question being asked or hypothesis being tested in this study?

Our previous research found that people report greater intentions to engage in risky driving behaviors if they are told they will be riding an autonomous vehicle system labeled Autopilot, rather than if the system is labeled Copilot. We also found indicative behavioral evidence for this effect – participants took longer to take control of a driving simulation in a risky scenario when it was labeled Autopilot rather than Copilot.

In this study, we aim to extend the behavioral evidence for this labeling effect. We ask the following -- In an incentive compatible setting, are people more likely to choose a system labeled Autopilot than Copilot in order to perform a complicated automated driving task, and are they more likely to feel comfortable watching the system's behavior when it is labeled Autopilot than Copilot?

3) Describe the key dependent variable(s) specifying how they will be measured.

The two main dependent variables are: (i) Whether people choose Autopilot or Copilot (categorical), and (ii) Comfort ratings after viewing the simulation (continuous).

4) How many and which conditions will participants be assigned to?

There are 2 conditions: simulation without notifications, and simulation with notifications.

In both conditions, participants are told they will see a driving simulation and then see one. In the "with notifications" condition, they are additionally told that they will see smartphone notifications pop up during the simulation and then they also see these notifications during the simulation.

We anticipate that participants will be even more likely to choose Autopilot (vs. Copilot) with notifications (vs. without); similarly, we anticipate them to feel even more comfortable with Autopilot (vs. Copilot) with notifications (vs. without).

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

We will run a binomial test against chance (50%) to test whether people choose Autopilot more than Copilot.

We will also run a logistic regression to test whether the proportion of participants choosing Autopilot varies as a function of the notification condition.

For comfort ratings, we will run a 2(label chosen: Autopilot, Copilot) x 2 (Notification: Present, Absent) ANOVA.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

We will exclude participants who fail either of two comprehension checks during the survey. We will also exclude Mobile users in our analysis, since the behavioral simulation is optimized for respondents using a Desktop.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

Informed by our prior research on driving simulation behavior, we plan to collect data from up to 2,000 participants following a Pre-Registered Interim Analysis Design (PRIAD; André and Reinholtz, 2024). We will conduct three interim analyses at sample sizes of 500, 1,000, and 1,500, followed by a final analysis at 2,000 observations. Applying the Pocock (1977) correction, data collection will conclude early if the primary analyses produce a p-value below 0.01821 at any interim stage; otherwise, recruitment will continue until the final sample size is reached.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

Only participants who correctly answer an attention check at the beginning of the survey will be eligible to participate.